Informer-Based LIBS for Qualitative Multi-Element Analysis of an Aceh Traditional Women's Medicine

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Abstract. Accurate elemental identification in Laser-Induced Breakdown Spectroscopy (LIBS) is hampered by dense line overlap, matrix effects, and the quadratic computational cost of standard self-attention on high-resolution spectra. We investigate an Informer-based deep learning model for qualitative multi-element LIBS analysis, trained entirely on a physics-based synthetic library generated from Saha-Boltzmann Local Thermal Equilibrium (LTE) plasma modelling. The Informer encoder employs ProbSparse self-attention, reducing the complexity to $\mathcal{O}(L \log L)$, and is optimized with a Multi-Label Focal Loss to handle class imbalance across 17 elemental labels. On 7,500 simulated test spectra, a compact 2-layer Informer attains a macro F_1 score of 0.64 (precision 0.59, recall 0.77), outperforming a deeper 3-layer variant that exhibited overfitting. As a proof of concept on experimental data, the trained model was applied to LIBS spectra of an Aceh Traditional Women's Medicine and successfully identified Na, Mg, Ca, and Mn in agreement with manual line identification using NIST Atomic Spectra Database. These results demonstrate that an Informer-based, simulation-trained LIBS model can transfer from synthetic to real spectra without empirical calibration standards and has practical potential as a rapid support tool for multi-element assessment of complex herbal preparations.

1. Introduction

Laser-Induced Breakdown Spectroscopy (LIBS) is a rapid optical emission technique for elemental analysis, requiring minimal sample preparation and capable of detecting multiple elements in a single measurement [1,2]. It is increasingly used in planetary exploration [3] and remote sensing [4] to biomedical applications [5], including the safety screening of traditional medicines [6]. In these contexts, fast and reliable elemental identification is essential for detecting contaminants, ensuring product quality, and supporting public health. Calibration-Free LIBS (CF-LIBS) approaches, in which line intensities are linked directly to plasma parameters and composition without empirical standards, aim to reduce the need for labour-intensive calibration and enable more flexible screening protocols [2,7]. In contrast to these quantitative CF-LIBS schemes, the present work addresses a qualitative, multi-label presence/absence task and adopts

a complementary simulation-trained strategy, using synthetic spectra from a physics-based plasma model to train an Informer network [8,9].

Despite these advantages, LIBS analysis remains challenging. Complex overlapping spectra arise because a single plasma shot produces a dense forest of emission lines from major and trace constituents, leading to substantial interference [6,10]. Plasma variability and matrix effects, driven by changes in temperature, electron density, and sample composition, introduce systematic biases that impede direct quantification [10]. Calibration-based LIBS requires extensive reference standards, whereas manual CF-LIBS (Saha–Boltzmann) relies on expert line identification and becomes unreliable when Local Thermal Equilibrium (LTE) or optically thin assumptions break down; both approaches demand high-quality data and still struggle with overlapping spectra [1]. To mitigate these limitations, recent studies have increasingly turned to machine learning for LIBS spectral analysis. Artificial Neural Networks (ANNs), Convolutional Neural Networks (CNNs), Bidirectional Long Short-Term Memory (Bi-LSTM) [11], and, more recently, transformer-based models have improved handling of overlapping features and matrix effects [4,12–15]. In particular, transformer architectures have achieved competitive performance in quantitative LIBS for rare-earth ores [12] and in high-speed LIBS classification for battery materials [14], while work on long biological sequences has emphasized that both tokenization and the efficiency of the underlying Transformer are critical for very long inputs [16,17].

Table 1. Concise comparison of prior works and the proposed Informer-based CF-LIBS framework

Approach / Methode	Application / Sample	Key Strength	Key Limitation
Ensemble CNN [13]	Planetary Exploration	High precision in capturing local spectral features for quantitative analysis.	Limited flexibility for varying input lengths; ignores global dependencies.
Bi-LSTM) [11]	Martian-like Minerals	Captures sequential dependencies better than standard NNs.	Sequential processing limits parallelization; computationally heavy for long spectra.
Deep Learning [15]	Industrial Aluminium Scrap	Robust quantitative regression for homogeneous industrial metals.	Requires extensive experimental datasets; less effective for qualitative biological screening.
Deep Neural Network [14]	Li-ion Battery Materials	Fast and accurate classification for fixed material classes.	Fully supervised; relies heavily on large, strictly labelled experimental datasets.
Physics-Informed [7]	Calibration & Background Removal	Unsupervised signal correction based on theoretical physics.	Focuses on signal preprocessing, not direct multi-element identification.
Informer (Proposed)	Aceh Traditional Women's Medicine	Linear complexity $\mathcal{O}(L \log L)$ for long sequences; Zero-Shot transfer from Physics-Based Simulation.	Currently optimized for qualitative screening, not yet for quantitative concentration.

These developments motivate the use of Informer, a long-sequence transformer that combines ProbSparse self-attention and sequence distilling to reduce the complexity of self-attention to $\mathcal{O}(L \log L)$ and to handle long one-dimensional sequences efficiently [9]. While deep learning approaches are gaining traction globally, applying them to heterogeneous tropical herbal matrices, as previously explored by our group on samples such as Betel Nut [6], remains a challenge due to the scarcity of labelled experimental datasets [18]. Existing LIBS studies have largely focused on Convolutional Neural Networks (CNNs) [13] and Recurrent Neural Networks (RNNs) [11], which do not scale well to high-resolution spectra with thousands of channels, and, to our knowledge, Informer or similar long-sequence transformers have not yet been applied to calibration-free LIBS for biological or herbal matrices. A concise comparison of these prior approaches and the proposed Informer-based framework is summarized in Table 1.

This leaves a gap for models that can perform high-resolution, calibration-free elemental identification in complex samples without relying on experimental calibration data. At the same time, traditional herbal medicines may accumulate potentially toxic metals from soil, irrigation water, or processing conditions, and their composition often reflects local agricultural and storage practices. Rapid, minimally destructive screening of such preparations is therefore relevant to environmental and public health monitoring [2]. By focusing on Aceh Traditional Women's Medicine as a case study, this work links advanced spectral modelling and long-sequence transformers to multi-element monitoring in heterogeneous plant-based materials. We develop an Informer-based deep learning model [9] trained entirely on a comprehensive, physics-based simulated spectral library [19], demonstrate its performance on both synthetic and experimental LIBS data, and discuss the implications for LIBS and related analytical techniques in complex herbal matrices.

2. Method

We developed a LIBS identification workflow that does not rely on empirical calibration standards, comprising three main stages: (1) generation of a large simulated spectral library using physics-based modelling, (2) design and training of an Informer-based deep learning model, and (3) validation on a real Aceh Traditional Women's Medicine sample. The task is formulated as a qualitative, multi-label classification of elemental presence or absence, rather than a quantitative estimation of concentrations.

Synthetic spectra were generated by combining tabulated atomic data from the National Institute of Standards and Technology (NIST) Atomic Spectra Database (ASD) [19] with a simple Local Thermal Equilibrium (LTE) plasma model. Level populations and ionization balance were obtained from Saha-Boltzmann equilibrium over a grid of plasma conditions covering electron temperatures (T_e) from 5,000 K to 15,500 K and electron densities (N_e) ranging from 1×10^{16} cm^{-3} to 1×10^{18} cm^{-3}, which encompasses the typical operating regime of our experimental setup [1]. Individual line profiles were modelled as Voigt functions that account for Doppler and collisional broadening and were convolved with an approximate instrumental response to emulate the finite spectral resolution of the LIBS system. A slowly varying continuum component representing bremsstrahlung, baseline effects were added, and the resulting spectra were sampled on a fixed 4096-channel grid covering the UV/VIS range from 200 to 900 nm.

Given the high-dimensional nature of LIBS spectra represented in this study as sequences of 4,096 channels, standard Transformer models suffer from quadratic computational complexity. To address this, we employed the Informer architecture, which utilizes a ProbSparse self-attention mechanism to achieve $\mathcal{O}(L \log L)$ efficiency [9]. The model is trained on 18 labels: 17

elements and 1 Background. The elemental classes (ordered by aggregate validation confidence in our runs) are: Ca, Si, Cr, Ni, Mg, Cl, Na, Mn, Al, S, Ar, O, C, N, Ti, Co, and Fe; the background class represents continuum and noise (bremsstrahlung, recombination, instrument baseline). For context, we group them as follows: major organics (C, N, O), minor minerals (Ca, Mg), trace metals (Al, Fe, Mn, Na), and additional atomic species (Si, Cr, Cl, Ni, Co, S, Ti, Ar).

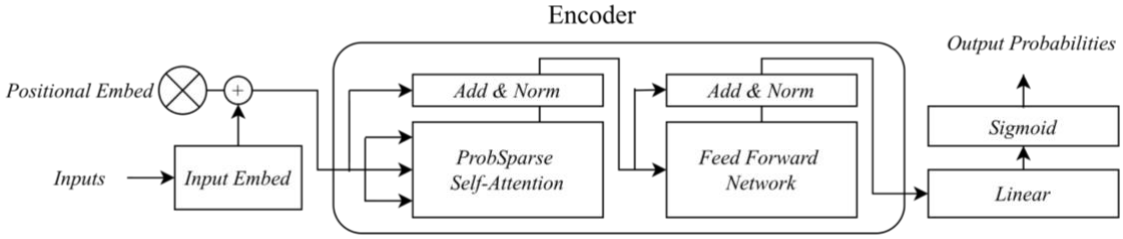


Figure 1. Modified Informer architecture for calibration-free LIBS (this study).

The Informer encoder used $d_{model} = 32$, $d_{ff} = 64$, and $H = 4$ attention heads with fixed positional encoding. Two encoder layers were retained as optimal for generalization. Training and optimization were performed on the Mahameru High-Performance Computing (HPC) system [20], utilizing the AdamW optimizer (learning rate 1×10^{-4}, weight decay 1×10^{-5}), dropout 0.4, batch size 26, up to 100 epochs with early stopping. MultiLabelFocalLoss ($\gamma = 2$) with class-weighted sampling and gradient accumulation equal to 4 mitigated class imbalance. A 2-layer architecture generalized best, whereas deeper 3-layer variants exhibited overfitting (F_1 score decreased from 0.64 to 0.52). Figure 1 schematically illustrates the Informer structure used in this study, adapted for full-spectrum (4096-channel) LIBS classification with a multi-label output head.

The proposed framework was validated using experimental LIBS spectra from an Aceh Traditional Women's Medicine sample. The sample, selected from 13 distinct traditional herbal formulations widely used in Aceh for postnatal care, was processed into a dried, homogenized powder to ensure uniformity and mitigate matrix heterogeneity. Laser ablation was performed using a Q-switched Nd:YAG laser (1064 nm, 10 Hz) operating at atmospheric pressure (1 atm). Plasma emission was collected and dispersed by an Optical Multichannel Analyser (OMA) coupled with an ICCD camera covering the UV–Vis range from 200 to 900 nm. To ensure data representativeness, spectra were acquired under fixed experimental conditions (energy, gate delay, and width) by averaging repeated shots on fresh sample positions. Manual line identification using the NIST Atomic Spectra Database [19] served as the ground truth for evaluating the Informer model's qualitative predictions.

3. Result and Discussion

The 2-layer Informer generalized best on simulated data (≈ 7,500 spectra), reaching a macro F_1 score = 0.6445 with precision = 0.5938 and recall = 0.7733, while a deeper 3-layer variant suffered from overfitting (F_1 score = 0.5199). Detailed per-element performance metrics comparing these architectures are provided in Table 2. Per-element trends mirror spectral physics: Ca, Cr, and Si performed strongly (F_1 score > 0.80), Al and Mg showed intermediate performance (with F_1 score ranging from 0.66 to 0.76), and Co and Fe performed lower (around 0.18 to 0.30) due to dense

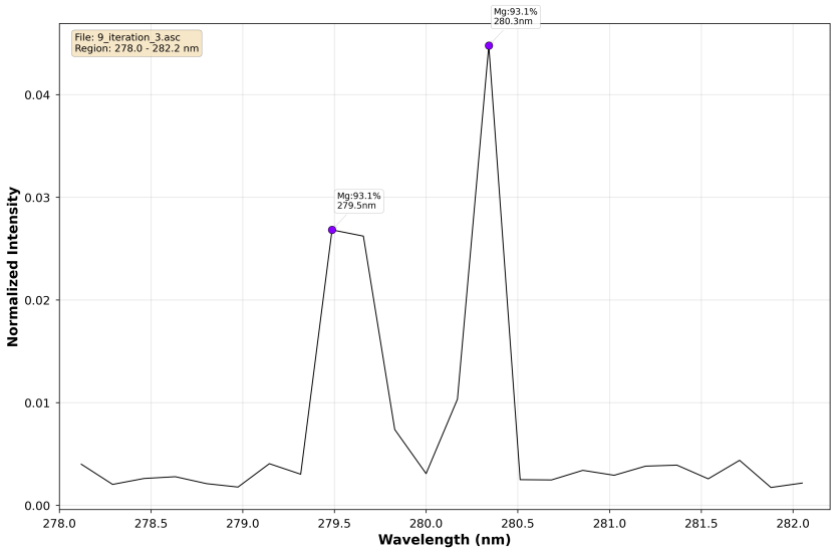
overlapping lines. These metrics define the model's optimal configuration before applying it to real LIBS data.

Table 2. Per-element precision (P), recall (R), and F_1 score (F_1) for the 2-layer vs 3-layer variant.

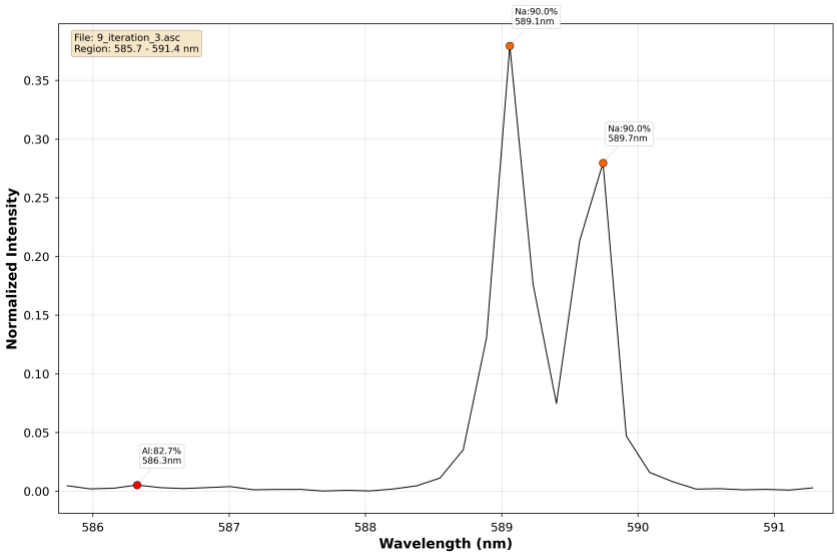
Species	P 2-Layer (%)	R 2-Layer (%)	F_1 2-Layer (%)	P 3-Layer (%)	R 3-Layer (%)	F_1 3-Layer (%)
Al	52.2	92.3	66.7	43.0	54.9	48.5
Ar	50.1	56.1	52.9	35.7	52.4	42.8
C	43.9	54.7	48.7	34.7	50.5	41.6
Ca	67.6	100.0	80.7	50.7	98.4	73.9
Cl	52.9	100.0	69.2	38.3	99.4	55.5
Co	55.2	20.2	29.5	50.7	18.1	27.7
Cr	72.5	99.7	83.9	51.1	97.3	72.1
Fe	57.8	10.4	17.6	62.9	15.1	24.2
Mg	62.5	98.1	76.4	45.4	96.0	61.7
Mn	75.0	88.7	81.3	60.4	88.2	72.0
N	51.0	65.0	57.2	34.1	64.2	44.7
Na	69.5	99.8	81.9	52.4	99.5	68.5
Ni	72.3	97.6	83.0	59.7	98.2	74.7
O	38.3	80.0	51.8	26.1	78.2	39.4
S	57.2	99.3	72.6	37.7	98.9	54.6
Si	72.4	100.0	84.0	41.8	99.7	58.7
Ti	58.8	50.5	54.3	56.2	49.2	52.5
Macro avg	59.4	77.3	64.5	46.7	65.8	52.0

Using the ≥ 0.60 confidence threshold defined in Methods, the experimental spectrum produced four positive predictions: Na, Mg, Ca, and Mn. All were consistent with manual/NIST verification and also observed in a companion measurement on the same sample (Figure 2). Qualitative conventional analysis (manual line identification with NIST ASD) confirmed all four elements; representative diagnostic lines are annotated in Figure 2: Mg (279.5, 280.3 nm) and the Na D-doublet (589.0 and 589.6 nm). The remaining elements were below the decision threshold. Table 4 summarizes the qualitative agreement between the Informer predictions and conventional LIBS analysis for the Aceh Traditional Women's Medicine sample.

Our Informer-based LIBS model, trained exclusively on a physics-based simulated spectral library and therefore not dependent on empirical calibration standards, demonstrated strong performance on both simulated and real Aceh Traditional Women's Medicine spectra. In particular, the optimized 2-layer Informer maintained a favorable balance between precision and recall on 7,500 simulated spectra and transferred this behavior to experimental data by correctly highlighting Na, Mg, Ca, and Mn in agreement with conventional LIBS analysis. This combination of physics-informed simulation and long-sequence attention, therefore, supports the use of Informer-based LIBS models as rapid, calibration-free screening tools for complex herbal matrices.



(a)



(b)

Figure 2. The markers (dots) denote predicted peaks that closely align with standard NIST atomic lines and manual inspection. The plots show key identified emissions in the herbal sample: (a) Mg lines at 279.5 and 280.3 nm, and (b) Na doublet at 589.0 and 589.6 nm.

The predicted elements agreed with an independent companion measurement on the same sample, which reported consistent qualitative composition: major organics (C, H, N, O), minor minerals (Ca, K, Mg, P), and trace metals (Al, Fe, Cu, Mn, Na, Rb, Si, Cr). The plasma parameters (electron temperatures ranging from 5,000 K to 15,500 K, the electron density ranging from $6 \times 10^{16}\ cm^{-3}$ $9.54 \times 10^{16}\ cm^{-3}$) also matched the simulated range, supporting the physical validity of the training setup.

Table 3. CF-LIBS Qualitative results for the Aceh Traditional Women's Medicine (companion study).

Element Group	Representative Elements	Qualitative Level
Biogenic light elements	C, H, O, N	very high
Nutritional minerals	Ca, K, Mg, P	high
Trace transition	Al, Fe, Cu, Mn, Na, Rb, Si, Cr	low (trace)

Beyond qualitative agreement, the companion study also performed a quantitative CF-LIBS analysis on multiple subsamples of the same preparation, providing ppm-level estimates for a broader set of elements. Those results confirm that the formulation is dominated by light biogenic elements and nutritional minerals, with potentially harmful transition metals present only at trace levels. In this context, the Informer-based model proposed here can be viewed as a rapid front-end for multi-element screening that precedes or complements more detailed quantitative analysis. Table 3 provides a compact qualitative summary of these CF-LIBS results by grouping elements into biogenic, nutritional, and trace-metal categories.

Table 4. Summary of elemental identification for the Aceh Traditional Women's Medicine sample. The Informer predictions are compared against conventional LIBS analysis

Element	Category	Informer Prediction	Conventional LIBS
C	Major organic	Below threshold	Detected
N	Major organic	Below threshold	Detected
O	Major organic	Below threshold	Detected
Ca	Minor mineral	Positive	Detected
Mg	Minor mineral	Positive	Detected
Na	Trace metal	Positive	Detected
Mn	Trace metal	Positive	Detected
Fe	Trace metal	Below threshold	Detected

Although the qualitative agreement is encouraging, the present model configuration also exhibits clear limitations. In particular, several elements that are known to be present at high or moderate levels in the companion CF-LIBS analysis (for example C, N, O and Fe) remain below the

decision threshold in the Informer predictions for the experimental spectrum. This reflects the fact that the current synthetic library does not yet capture all relevant molecular bands, self-absorption effects, and variations in plasma conditions that occur in real herbal matrices. Addressing these limitations will require expanding the physics-based simulator to include molecular emission and non-LTE effects, as well as refining the decision thresholding strategy or calibration of predicted probabilities for experimental data.

4. Conclusion

We developed a physics-informed, calibration-free LIBS framework using an Informer Transformer trained solely on simulated Saha-Boltzmann spectra. The optimized 2-layer model achieved a macro F_1 score of 0.64, outperforming deeper baselines, and successfully identified Na, Mg, Ca, and Mn in Aceh Traditional Women's Medicine without empirical calibration. This validates the Informer's capability to generalize from physical simulations to complex experimental matrices. Future work will extend the simulated library to include molecular emissions and non-LTE effects to further enhance sensitivity for organic-rich samples.

Acknowledgments

The authors gratefully acknowledge financial support from the Universitas Syiah Kuala Master's Thesis Research Program under Contract No. 404/UN11.LI/PG.01.03/14557-PTNBH/2025. We thank Universitas Syiah Kuala and the National Research and Innovation Agency (BRIN) for their institutional support. Computational resources were provided by the Mahameru High-Performance Computing (HPC) system at BRIN.

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